



Bellabeat Case Study

How can a wellness company play it smart?

By: Pratik Vindyala | Date: July 25th, 2026

Table of Contents

06

About Bellabeat

Company, products, and business goal

08

Ask

Business task at hand

07

Methodology

Framework & tools used

09

Prepare

Factors for choosing data

Table of Contents Cont.

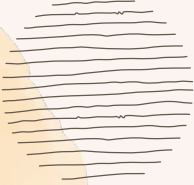
10 **Process**
Cleaning & data integrity

11-15 **Analyze**
Key findings

16 **Share**
Data visualization via
Dashboard

17 **Act**
Changes to implement

Table of Contents Cont.



18-20

Recommendations

Solutions based on my
analysis of Bellabeat

21

Conclusion

Summary of overall case
study

22

Sources

References used
throughout presentation

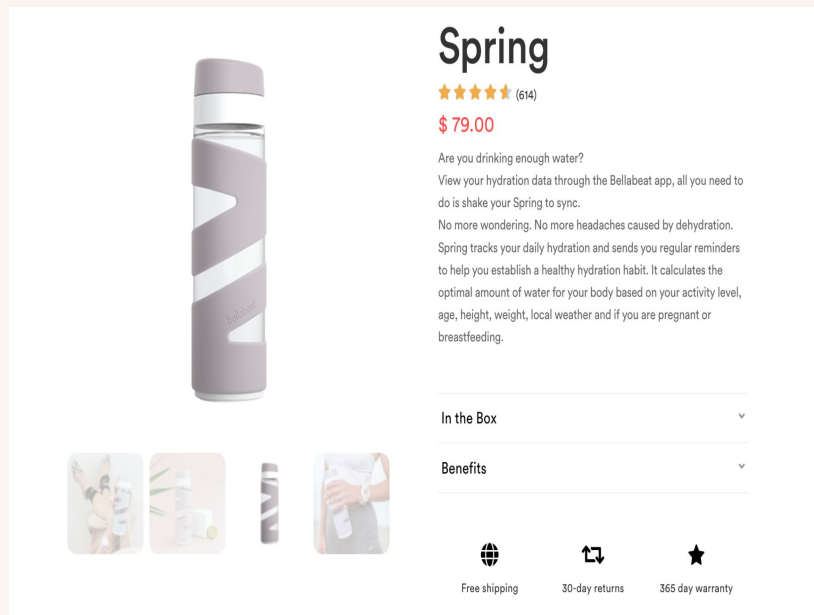


Background



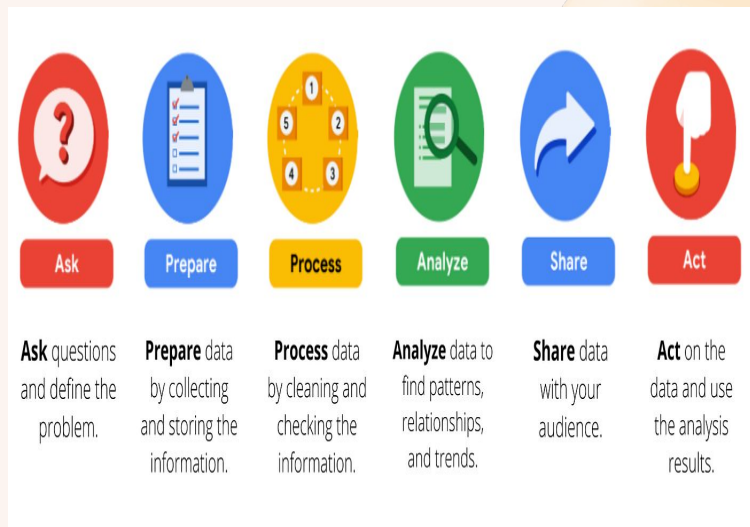
About Bellabeat

- Founded 2013, women's wellness
- Digital-first: Google, Meta, YouTube
- Products: App, Leaf, Time, Spring
- Goal: Data unlocks growth



Methodology & Tools

- Framework: Ask → Prepare → Process → Analyze → Share → Act
- Tools used
 - Google Sheets - Data cleaning & analysis
 - SQL (Big Query) - Data cleaning & analysis
 - Tableau - Data Visualization



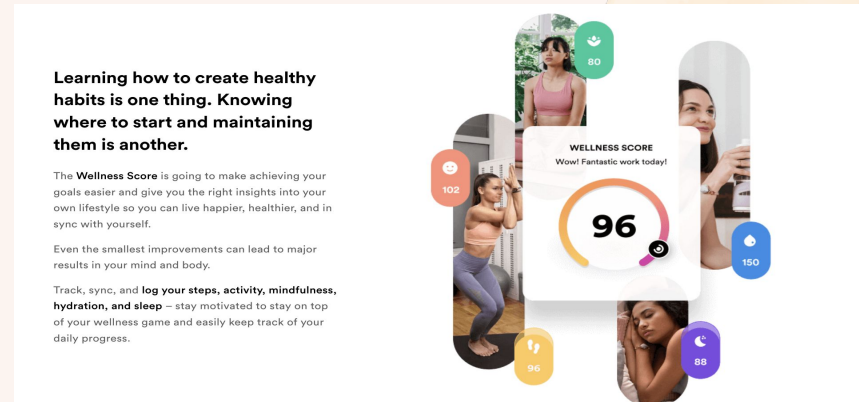
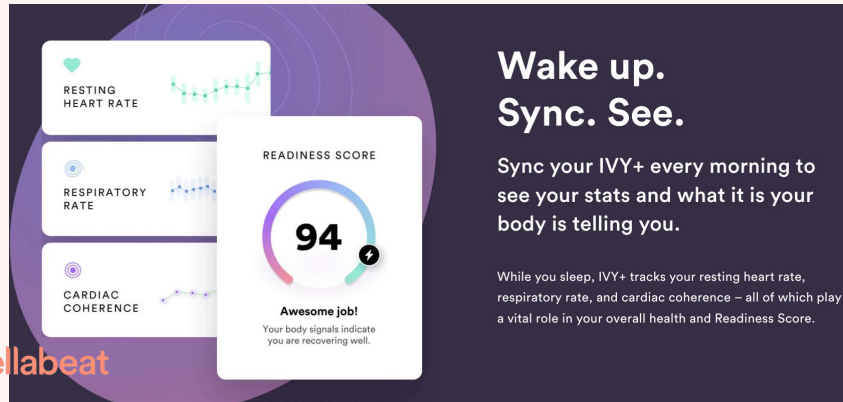


Data Analysis Process



Phase 1: Ask

- Business task: How should Bellabeat app build lasting habits
- Stakeholders: Sršen, Mur, and marketing analytics team
- Guiding questions: Problems & business decisions





Phase 2: Prepare

- FitBit Fitness Tracker Data (primary)
- 35 users, 11 files, ~30 days
- Limitations: mall sample & no gender field
- WHO Global Health Observatory (supplementary)
 - Chosen → Solves biggest issue with primary data



World Health
Organization

Phase 3: Process

- Clean: removed 525 duplicate rows
- Scale: 5 files over 1.1M rows → BigQuery
- Verify: cross-checked BigQuery vs. Sheets
- Identifying right tool for data size
 - Google Sheets vs. BigQuery

THE DATA CLEANING PROCESS STEP BY STEP



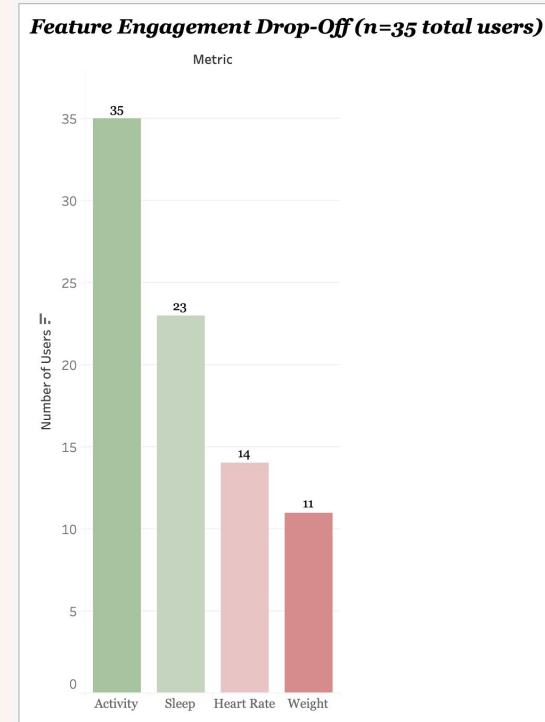
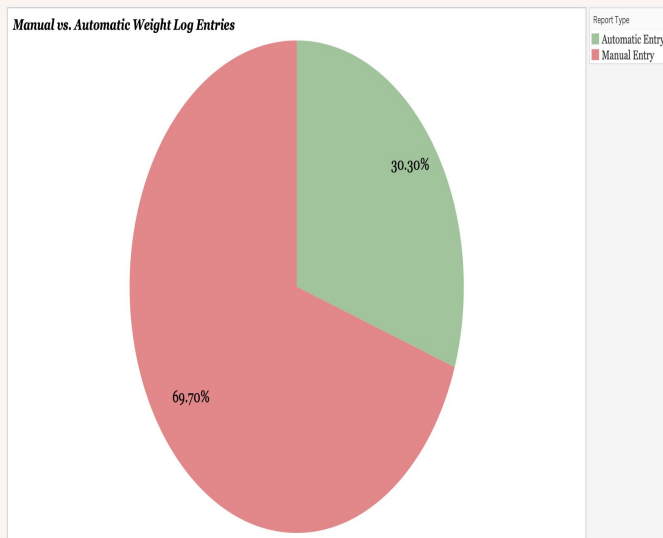


Phase 4: Analyze

- Three key questions
 - Q1 → Engagement: equal tracking or drop-off
 - Q2 → Timing: when are people active?
 - Q3 → Relationship: activity predict sleep vs. WHO?

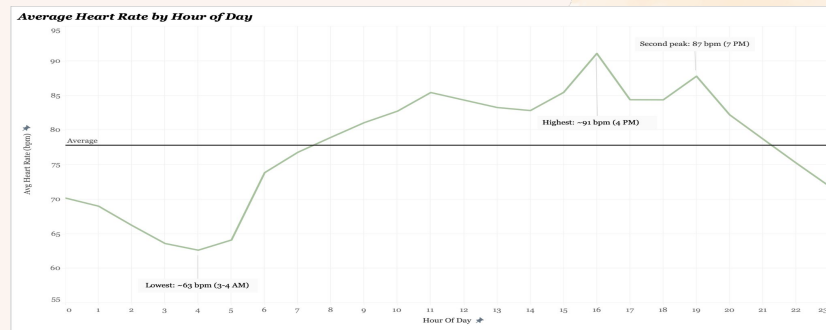
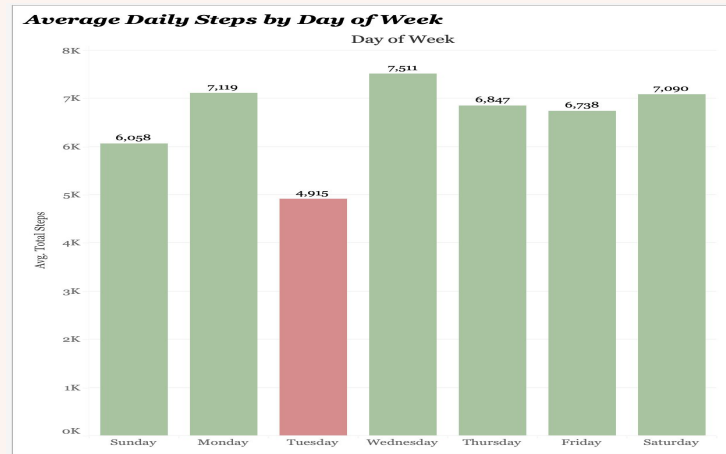
Finding #1: The Consistency Gap

- Activity: 35 users
- Sleep: 23 users
- Heart Rate: 14 users
- Weight: 11 users
- 69.7% of weight entries manual



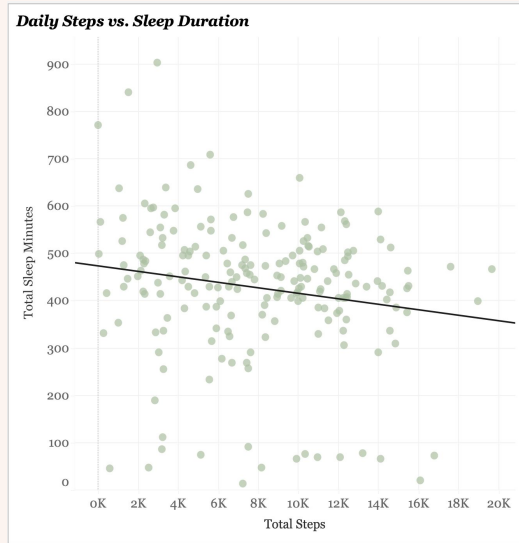
Finding #2: Activity Has a Rhythm

- Heart rate lowest: ~63 bpm (3–4 AM)
- Sleep: 23 users
- Heart Rate: 14 users
- Weight: 11 users
- 69.7% of weight entries manual

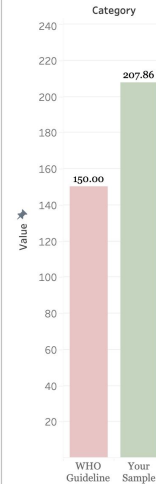


Finding #3: Active, But Sleep Differs

- Sample: 207.9 min/week active
- WHO guideline: 150 min/week
- +38% above minimum
- Steps vs. sleep: $R^2 = 0.03$
- Statistically real, practically negligible

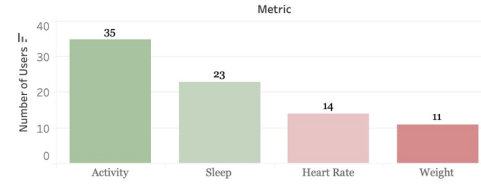


Sample Exceeds WHO Weekly Activity Guideline

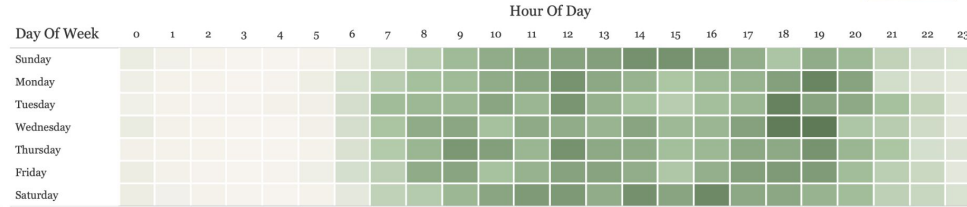


Bellabeat App: The Consistency Gap

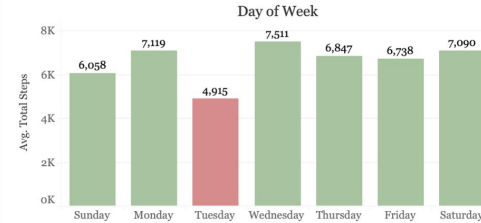
Feature Engagement Drop-Off (n=35 total users)



Activity Intensity by Day and Hour



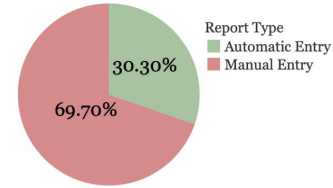
Average Daily Steps by Day of Week



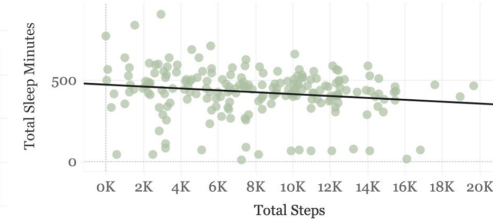
Sample Exceeds WHO Weekly Activity Guideline



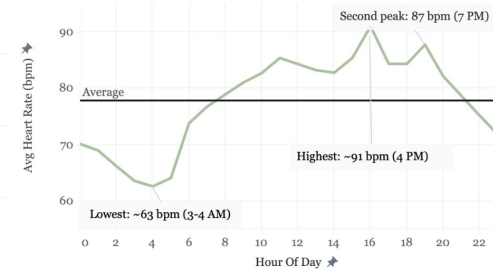
Manual vs. Automatic Weight Log Entries



Daily Steps vs. Sleep Duration

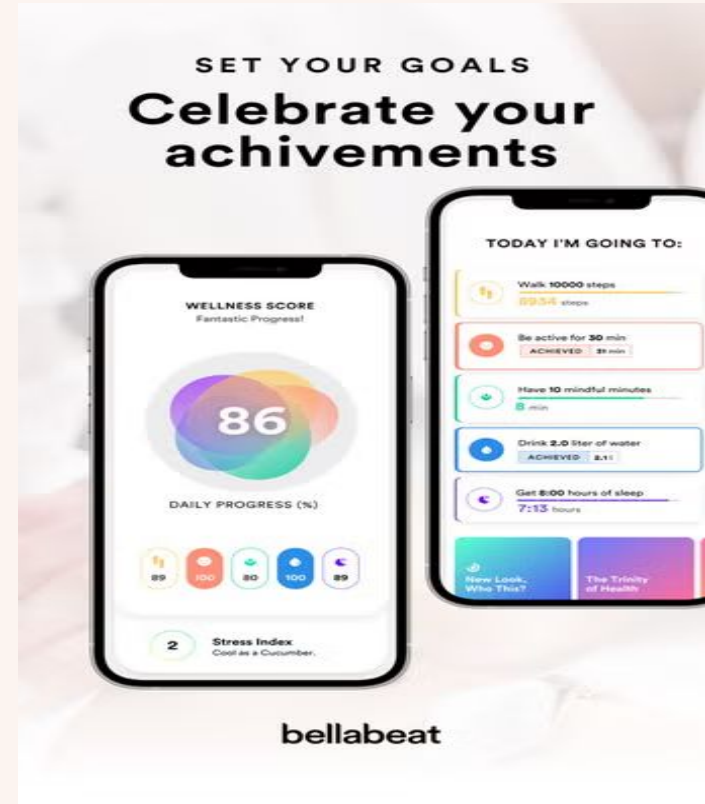


Average Heart Rate by Hour of Day



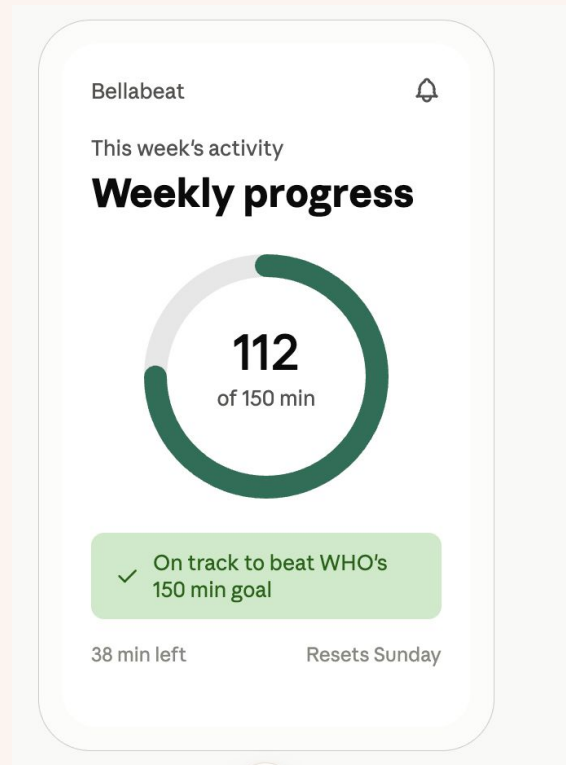
Phase 5 Act: Overview

- Reframe tracking as goal-oriented
- Time nudges to real behavior
- Lower friction on weak spots



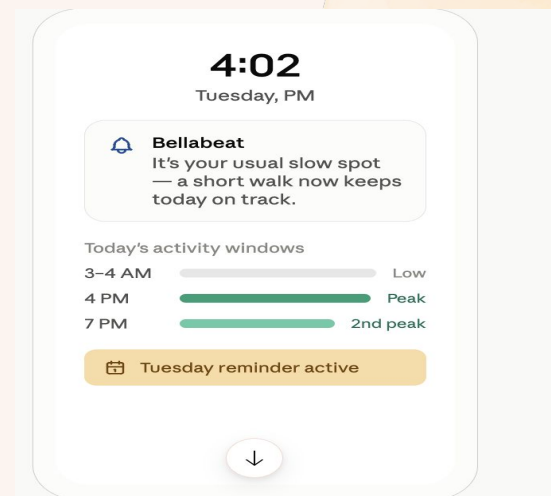
Recommendation 1: Show Progress Toward WHO Guideline

- Evidence: sample exceeds WHO by 38%
- Recommendation: in-app weekly progress indicator
- Why: turns tracking into a target



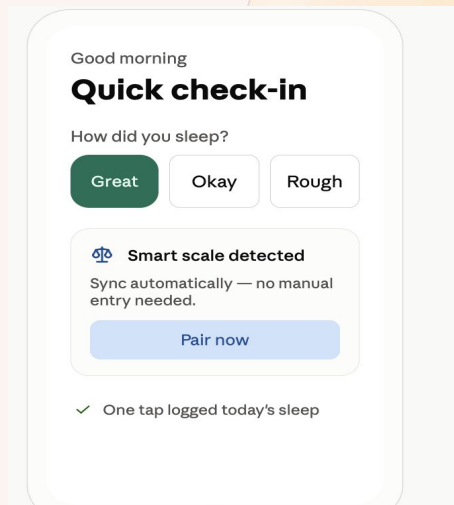
Recommendation 2: Time Nudges to Real Behavior

- Evidence: peaks at 4 PM & 7 PM; Tuesday dips 23%
- Recommendation: schedule nudges to peak hours & Tuesday reminder
- Why: matches real patterns, not guesses



Recommendation 3: Lower Friction on Weakest Metrics

- Evidence: 69.7% weight entries manual; only 11 of 35 log weight
- Recommendation: prioritize low-friction logging for weight & sleep
- Why: biggest gap has most headroom



Conclusion

- Sample exceeds WHO guideline by 38%
- Opportunity: consistency, not activity
- Next: test, re-measure, expand data



Activity
35 of 35 users
100% consistent



Sleep
23 of 35 users
66% consistent



Heart rate
14 of 35 users
40% consistent



Weight
11 of 35 users
31% consistent

🎯 The goal: bring all four rings to 100%, not just activity



Sources

- Kaggle
- WHO



Thank You

